

SCH3U - Gas Unit Review

Useful Information/Equations that you need to know:

- Pressure Conversion: $101.3 \text{ kPa} = 1 \text{ atm} = 760 \text{ mmHg}$
- Temperature Conversion: $^{\circ}\text{C} + 273 = \text{K}$
- STP = 22.4 L/mol (molar volume of a gas), 273 K , 101.3 kPa
- SATP = 24.8 L/mol (molar volume of gas), 298 K , 100 kPa
- $R = 8.314 \text{ kPaL/molK}$

Boyle's Law

Charles' Law

Gay-Lussac's Law

Combined Gas Law

Dalton's Law of Partial Pressures

Avogadro's Law

Ideal Gas Law

Chapter 11 - Behavior of Gases

- 11.1 - Kinetic Molecular Theory, intermolecular forces, characteristics of an "ideal" gas
- 11.2 - what is "pressure", atmospheric pressure, relationship between P and V (Boyle's Law)
- 11.3 - temperature and volume relationship (Charles' Law), absolute zero, relationship between pressure and temperature (Gay-Lussac's Law)
- 11.4 - STP vs SATP (conditions), combined gas law, partial pressures (Dalton's Law of Partial Pressures)

Chapter 12 - Exploring Gas Laws

- 12.1 - law of combining volumes and Avogadro's Law, molar volume of gas at STP and SATP, "real" gases, ideal gas law
- 12.3 - stoichiometry problems: volume to volume, mass to volume, vapour pressure/dry gas

Good Review Questions : (answers on pages 645-646)

Chapter 11 Review - Page 467-469

1, 2, 4, 12, 13, 16 - 26, 28, 31

Chapter 12 Review - Page 521-523

4, 5, 6, 12, 14, 16, 17, 20 - 22, 26 - 30

Unit 4 Review - Page 526-528

1-5, 7-9 (MC), 13, 15 - 24, 26-35